

WEIPA, Australia

WMO: 941700

Lat: 12.6778S

Lon: 141.9208E

Elev: 19

StdP: 101.10

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	17.8	9.8	7.6	30.2	11.6	8.5	29.2	9.8	29.1	9.0	28.9	1.2	130	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.0	35.8	22.6	34.9	22.7	34.1	22.9	27.0	30.5	26.7	30.1	26.4	29.8	4.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	28.4	25.8	21.1	28.0	25.6	20.8	27.8	85.3	30.6	83.9	30.4	82.7	29.9	36.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.6	7.0	DB	14.3	37.5	1.7	0.7	13.1	38.0	12.1	38.4	11.1	38.7	9.9	39.2	(4)
(5)			WB	12.8	29.0	1.6	2.6	11.7	30.9	10.8	32.4	9.9	33.9	8.8	35.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	27.8	27.6	27.5	27.3	26.3	25.2	24.6	25.0	26.7	28.3	29.1	28.8	(6)	
	DBStd	1.85	1.17	1.12	0.96	0.93	1.14	1.34	1.38	1.31	1.27	1.23	1.00	1.14	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6207	551	492	544	518	506	454	452	466	500	567	572	583	(10)	
	CDD18.3	3165	293	259	286	268	248	204	194	208	250	308	322	325	(11)	
	CDH23.3	30748	2864	2438	2633	2551	2202	1608	1547	1843	2503	3363	3656	3541	(12)	
(13)	CDH26.7	12107	975	771	859	936	777	505	483	698	1158	1642	1763	1541	(13)	
(14)	Wind	WSAvg	3.1	2.6	2.5	2.4	3.1	3.5	3.6	3.7	3.5	3.5	3.0	2.6	(14)	
(15) Precipitation	PrecAvg	1883	470	494	399	105	15	3	1	1	4	22	110	260	(15)	
	PrecMax	2690	1112	919	1063	352	129	32	10	10	33	116	451	531	(16)	
	PrecMin	1208	182	159	73	1	0	0	0	0	0	0	0	41	(17)	
	PrecStd	390	198	199	199	93	22	6	2	2	6	29	90	119	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	33.5	33.2	33.2	32.7	32.1	32.0	33.2	35.4	36.7	36.9	36.3	(19)	
		MCWB	24.8	25.4	25.2	24.7	24.0	22.4	21.3	20.7	21.0	22.2	22.4	23.8	(20)	
	2%	DB	32.9	32.4	32.4	32.4	31.8	31.2	31.0	32.2	34.3	35.6	35.9	35.0	(21)	
		MCWB	25.5	25.7	25.2	24.4	23.4	22.2	21.2	20.7	21.2	22.2	22.6	24.1	(22)	
	5%	DB	31.9	31.6	31.7	31.8	31.2	30.4	30.2	31.3	33.3	34.6	34.9	33.8	(23)	
		MCWB	25.7	25.9	25.3	24.1	23.1	22.1	20.9	20.6	21.2	22.1	22.8	24.6	(24)	
	10%	DB	31.0	30.7	30.9	31.1	30.5	29.6	29.4	30.4	32.3	33.6	33.8	32.8	(25)	
		MCWB	25.9	25.9	25.5	24.1	23.0	21.8	20.7	20.5	21.2	22.1	23.3	25.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	27.2	27.2	26.7	25.8	24.9	24.0	24.1	24.9	26.2	26.9	27.2	(27)	
		MCDB	30.6	30.6	30.5	29.9	29.7	28.5	28.3	28.7	29.3	30.2	31.0	30.9	(28)	
	2%	WB	26.9	26.7	26.7	26.1	25.1	23.9	23.2	23.2	24.1	25.3	26.2	26.7	(29)	
		MCDB	30.2	30.0	29.9	29.5	29.1	28.3	27.7	28.2	28.9	29.5	30.2	30.6	(30)	
	5%	WB	26.6	26.5	26.3	25.7	24.6	23.2	22.5	22.6	23.6	24.7	25.8	26.4	(31)	
		MCDB	29.8	29.6	29.4	29.1	28.6	27.8	27.4	27.8	28.6	29.3	29.8	30.2	(32)	
	10%	WB	26.2	26.2	26.0	25.3	24.1	22.7	21.9	22.0	23.0	24.2	25.4	26.1	(33)	
		MCDB	29.3	29.1	29.1	28.5	28.2	27.4	27.1	27.4	28.3	29.2	29.4	29.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.3	6.9	8.3	9.2	9.8	10.8	12.1	13.0	12.4	11.0	8.8	(35)	
		MCDBR	8.3	7.7	8.3	9.3	9.9	10.6	11.6	12.9	13.9	13.3	12.5	10.5	(36)	
	5% WB	MCWBR	2.6	2.5	2.6	2.8	3.1	3.2	3.5	3.9	3.9	3.6	3.4	3.0	(37)	
		MCWBR	6.8	6.6	6.9	7.5	8.2	9.1	9.7	11.2	12.4	11.6	9.9	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.409	0.406	0.392	0.389	0.376	0.369	0.360	0.374	0.395	0.407	0.418	0.409	(40)	
		taud	2.519	2.528	2.570	2.555	2.567	2.574	2.588	2.528	2.452	2.426	2.424	2.489	(41)	
	Ebn at Noon		935	931	926	896	877	867	885	898	909	918	920	933	(42)	
		Edh at Noon	113	111	104	101	94	90	91	102	115	122	124	116	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.11	5.16	5.18	5.37	4.93	4.52	4.77	5.56	6.41	6.91	6.84	6.09	(44)	
	RadStd		0.61	0.82	0.41	0.45	0.29	0.31	0.26	0.32	0.27	0.34	0.40	0.66	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	+112	+109	(46)	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	+112	+109	(47)	

Nomenclature: See separate page